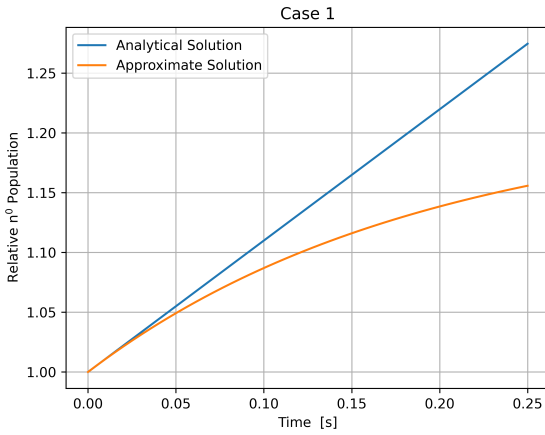


NE 511: Homework 2

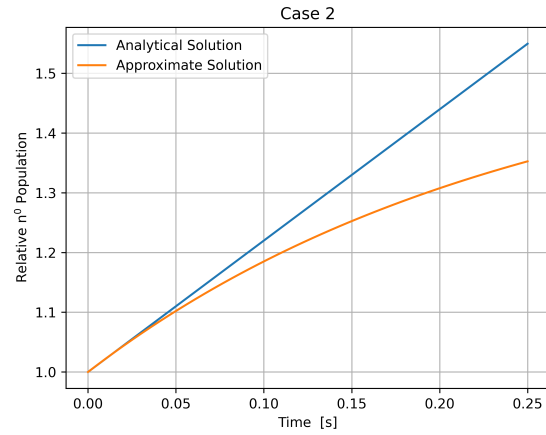
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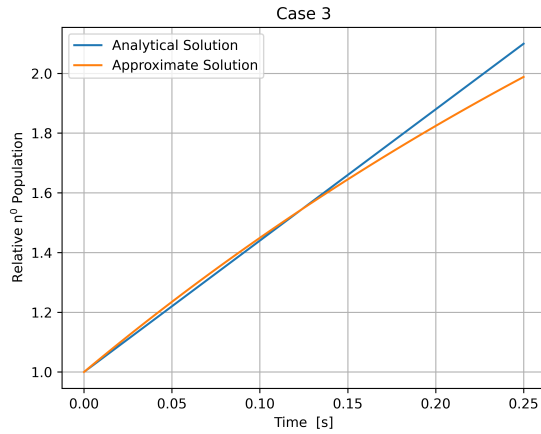
1 Analytical vs Approximate One-Group PKE



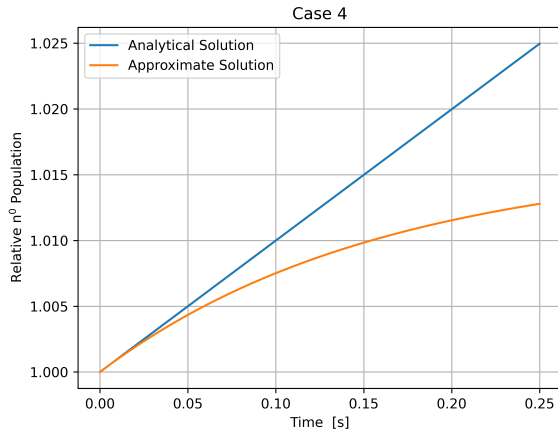
(a) Case 1, $\rho = 0.0011$



(b) Case 2, $\rho = 0.0022$



(c) Case 3, $\rho = 0.0044$



(d) Case 4, $\rho = 0.0001$

Looking at all the plots, we only change ρ and we see the higher the ρ , the closer the agreement with the analytical solution. We assume we will see closer agreement with a larger reactivity insertion. This makes sense according to our assumption that $\Lambda\lambda \ll \rho$ – this assumption fails when ρ is too small (e.g Case 4).

2 Prompt Burst Width

1. First, find the initial prompt reactivity using Eq. II.78.

$$\rho_{p1} = \rho_1 - \beta = 1.1\$ - 1\$ = 0.1\$ = 0.00075 \quad (1)$$

2. We will calculate the using the relative power assuming $p_0 = 1$. Then, use Eq. II.80 to find p^0 .

$$p^0 = \frac{\rho_1}{\rho_{p1}} p_0 \quad (2)$$

3. Additionally, we will use $\gamma = -0.8\$/\text{fp-s}$ and $\gamma_e = 0.006\Delta k/k/\text{fp-s}$, which are typical of an LWR. Using Eq. II.95

$$p_m = p^0 - \frac{\rho_{p1}^2}{2\Lambda\gamma_e} \quad (3)$$

4. Then, use Eq. II.110.

$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \left(1 - \frac{p^0}{p_m} \right) = 44.9\text{ms} \quad (4)$$

5. Next, we want to calculate the energy released using the given formula and Eq. II.109. Making sure to use ρ_{p1} in units of \$.

$$Q(t_2) = \int_0^{t_2} P(t)dt = -\frac{2\rho_{p1}}{\gamma} = 0.25 \text{ fp-s} \quad (5)$$

2.A Neglecting Heat Release

1. We start with the λ_h for a 0.6 cm diameter oxide fuel rod from the notes.

$$\lambda_h \approx 0.5 \frac{1}{\text{s}} \quad (6)$$

2. We can compare the burst width and the heat transfer time by multiplying the two in an exponent.

$$\exp(\lambda_h \Delta t) - 1 = 2.27\% \quad (7)$$

3. Neglecting heat release is a good assumption as 2.27% is very low considering the uncertainties involved.

2.B Neglecting P_0

1. To find the cooling by neglecting P_0 , we evaluating the following integral. Keeping in mind we assumed

$$P_0 = p_0 = 1$$

$$\int_0^{\Delta t} dt P_0 = P_0 \Delta t = 0.04488 \text{ fp-s} \quad (8)$$

2. We compare the value of the integral to the $Q(t_2) = 0.25 \text{ fp-s}$ from before to find the ratio of neglected cooling to total burst energy.

$$\frac{p_0 \Delta t}{Q(t_2)} = 0.17954 = 17.954\% \quad (9)$$

3. We see neglecting the cooling is not a good assumption – 17.954% is too high.

3 Sub-prompt to Prompt Critical Jump

We will use Eq. II.61. γ is given in units of $\frac{\$}{\text{fp-s}}$, so $\beta = \$$ for both cases.

$$\frac{\gamma}{\beta} = -\frac{\bar{\lambda}}{p^{00}} \Rightarrow p^{00} = -\frac{\bar{\lambda}\beta}{\gamma} \quad (10)$$

3.A Thermal U-235 Fueled Core

$$p^{00} = -\frac{\bar{\lambda}_{U235}\beta_{U235}}{\gamma_{U235}} = 2.5 \quad (11)$$

3.B Fast MOX Fueled Core

$$p^{00} = -\frac{\bar{\lambda}_{MOX}\beta_{MOX}}{\gamma_{MOX}} = 7.5 \quad (12)$$

The p^{00} values are lower than the SEFOR experiments because we assumed γ is roughly 8 times as high for the thermal U235 fueled core and 4 times higher for the MOX core compared to the calculated γ of $-0.022\$/\text{fp-s}$. In the real experiment, the delayed neutrons precursors are not omitted, which could be a reason for the higher peak power.

4 Super-prompt Critical Step Insertion

1. In this problem, we are solving for the relative power amplitude evolution over time using the following formula using Eq. II.107. The reference numbers for the following equations are referring to the documents "C2 Short-Time Multi-Physics Phenomena in Reactor Core (Part 1,2,3)."

$$p(t) = \frac{p_m}{\left\{ \cosh \left[\frac{\rho_b}{2\Lambda} (t - t_m) \right] \right\}^2} \quad (13)$$

2. We are given the following:

$$p_0 = 0.1 p_n \quad (14a)$$

$$\gamma = -0.08 \frac{\$}{\text{fp-s}} \quad (14b)$$

$$\frac{\beta}{\Lambda} = 0.62 e4 \frac{1}{\text{s}} \quad (14c)$$

$$t_m = 16.5 \frac{\Lambda}{\rho_{p1}} \quad (14d)$$

3. We will also need the following assumptions. We will assume $p_n = 1$ and compare everything against the nominal power level.

$$p_n = 1 \quad (15)$$

4. We will also assume $\rho_1 = 1.1\$$, which is the same as the derivation in the notes.

$$\rho_1 = 1.1\$ \quad (16)$$

5. We do not need to assume a value for β and will use the fact that $\beta = 1\$$.

$$\beta = 1\$ \quad (17)$$

6. We can calculate Λ based on the given relationship.

$$\frac{\beta}{\Lambda} = 0.62 e4 \frac{1}{\text{s}} \Rightarrow \Lambda = \frac{\beta}{0.62 e4} \text{s} \quad (18)$$

7. We need Eq. II.75.

$$p(0) = p^0 = \frac{\rho_1}{\rho_1 - \beta} p_0 \quad (19)$$

8. We need Eq. II.78.

$$\rho_{p1} = \rho_1 - \beta \quad (20)$$

9. We need Eq. II.95.

$$p_m = p^0 - \frac{\rho_{p1}^2}{2\Lambda\gamma} \quad (21)$$

10. We need Eq. 100b.

$$\rho_b^2 = \rho_{p1}^2 - 2\Lambda\gamma p^0 \quad (22)$$

11. In all the the above equations, the parameters are defined such that:

- (a) $p(t)$ is the relative power amplitude over time and what we are solving for,
- (b) p_n is the nominal power,
- (c) p_m is the maximum power amplitude during the transient,
- (d) ρ_b is the effective initial prompt reactivity given by the Nordheim-Fuchs model,
- (e) Λ is the mean neutron generation time,
- (f) t is the time,
- (g) t_m is the time at which the maximum power (p_m) occurs,
- (h) γ is the reactivity coefficient accounting for Doppler broadening and thermal expansion,
- (i) β is the delayed neutron fraction,
- (j) ρ_1 is the initial step reactivity,
- (k) ρ_{p1} is the initial prompt reactivity,
- (l) p_0 is the power at $t = 0$,
- (m) p^0 is the psuedo-initial power.

12. Then, we can plot

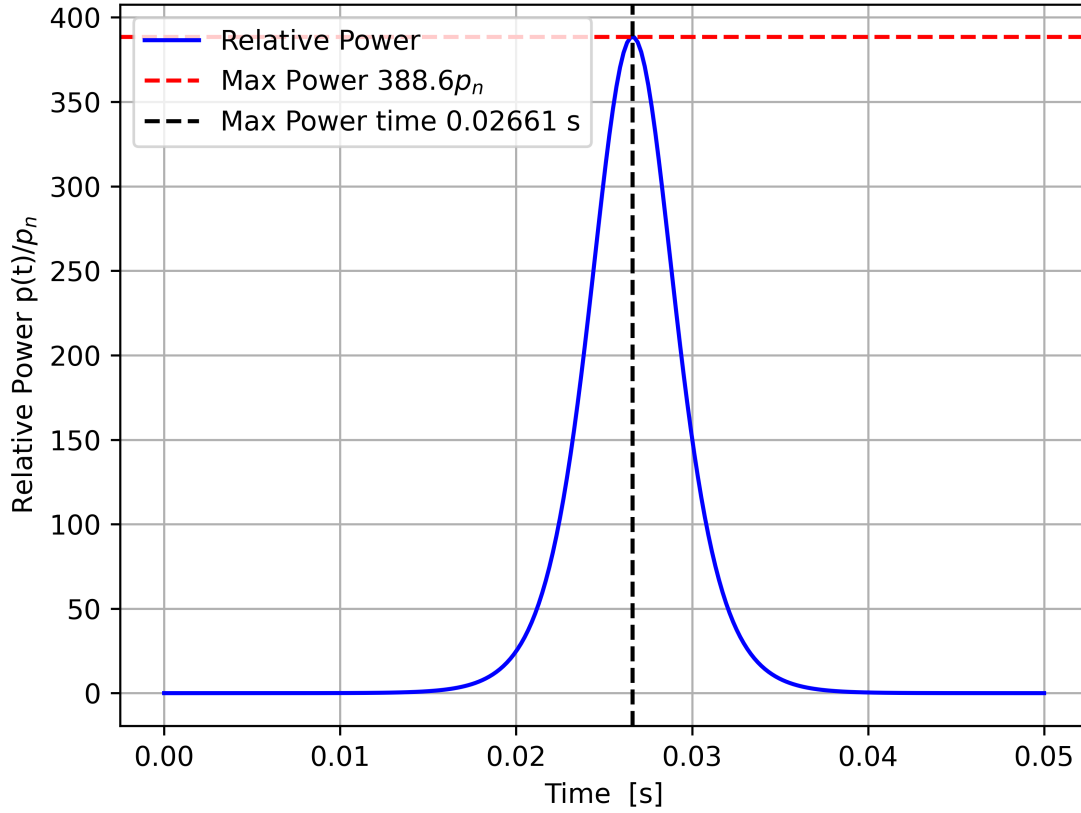


Figure 1: Relative Power Over Time

13. About the max power time, the relative power looks like Gaussian peak ramping from $0.1p_n$ at $t = 0$ to a massive peak of $388.6p_n$ at the maximum power time. The power then decreases as it increased to nominal power after the transient. This peak is massive and would surely cause issues if the power level was anything substantial.